

CE 310 — Week 7 In-Class Exercise: Correlation — Covariance, Pearson r, and Scatter Plots

Dataset: EnergyPlus DOE Reference Medium Office — Tucson TMY3, 2023 — Week5_D0E_Med0ffice_ZoneTemp_2023.csv (8,760 rows) **Topic:** CORREL · COVARIANCE.S · r^2 · conditional correlation · scatter plots

Before You Begin

- Download Week5_D0E_Med0ffice_ZoneTemp_2023.csv from D2L before the in-class exercise (8,760 rows, one per hour).
- Download the Wk07.04_In_Class_Answer_Sheet.xlsx from D2L before the in-class exercise. All answers go there.
- Column reference: G = Zone_Temp_F, H = Outdoor_Temp_F, I = Cooling_kWh.
- Verify your row count: =COUNTA(A2:A8761) should return 8760.

Answer Sheet column map:

Column	What to enter
Col A	(pre-filled question # — do not edit)
Col B	(pre-filled question text — do not edit)
Col C	Your Excel formula or method description
Col D	★ Intermediate value (★ questions only — leave blank otherwise)
Col E	Your numerical answer
Col F	Unit (pre-filled — do not edit)

Your answer cell matches the question label — A1 in row A1, B3 in row B3.

Type your formula as text. In Col C, put an apostrophe before the = so Excel stores the formula instead of running it — for example, '=AVERAGE(B2:B8761)'. Without the apostrophe Excel evaluates it and your method is lost. **Screenshots are not accepted;** the typed formula is what earns the approach credit.

Excel Quick Reference

Task	Formula
Pearson r (all rows)	=CORREL(H2:H8761, I2:I8761)
Sample covariance	=COVARIANCE.S(H2:H8761, I2:I8761)
Conditional r — FILTER	=CORREL(FILTER(H:H, E:E="Summer"), FILTER(I:I, E:E="Summer"))
Conditional r — Array (Ctrl+Shift+Enter)	{=CORREL(IF(E2:E8761="Summer", H2:H8761), IF(E2:E8761="Summer", I2:I8761))}
Count subgroup size	=COUNTIF(E2:E8761,"Summer") or =COUNTIF(F2:F8761,"Yes")

Task	Formula
r^2 (coefficient of determination)	= r_cell ^ 2
Standard deviation (sample)	=STDEV(I2:I8761)
Verify: $r = \text{cov} / (\sigma_1 \times \sigma_2)$	= COVARIANCE.S(H2:H8761,I2:I8761) / (STDEV(H2:H8761) * STDEV(I2:I8761)) should equal CORREL

Background

This week you'll quantify how strongly each variable moves with cooling energy.

Setup

You do all of this week's work in **one file: your Answer Sheet**. The data, your scratch work, and your charts all live there, so everything you built is in the file you hand in. Add as many sheets as you like — only Instructions, Answers, Charts and Written have to keep their names.

1. Open your Answer Sheet, add a sheet named Data, and import Week5_D0E_MedOffice_ZoneTemp_2023.csv onto it.
2. Verify 8,760 data rows: =COUNTA(A2:A8761) → should return 8760.
3. Confirm column letters: A = Date, B = Month, C = Hour, D = DayType, E = Season, F = BusinessHours, G = Zone_Temp_F, H = Outdoor_Temp_F, I = Cooling_kWh.
4. On the Answers sheet, enter your name and NetID.

Section A — Introduction to CORREL (8 questions × 2 pts = 16 pts)

Section A introduces correlation as a measurement: a single number, computed across the full year, describing how tightly two variables move together.

Each question uses CORREL or COVARIANCE.S on the full 8,760-row dataset, or simple arithmetic on those results. Enter answers rounded to 4 decimal places.

Formula syntax: =CORREL(array1, array2) — returns Pearson r between two arrays. =COVARIANCE.S(array1, array2) — returns sample covariance (ddof = 1).

A1 (2 pts) — r(Zone Temp, Outdoor Temp)

Compute r(Zone_Temp_F, Outdoor_Temp_F) for all 8,760 hours.

Moderate-to-strong — but note the 5.8× compression ratio, $\sigma_{\text{outdoor}} / \sigma_{\text{zone}} \approx 16.92 / 2.92$: the HVAC holds the zone far steadier than the weather outside it.

A2 (2 pts) — r(Outdoor Temp, Cooling): The Main Pair

Compute r(Outdoor_Temp_F, Cooling_kWh) for all 8,760 hours.

This is the main pair for this in-class exercise. A strong positive correlation — hotter outdoor hours tend to have higher cooling loads.

A3 (2 pts) — $r(\text{Zone Temp, Cooling})$: The Weak Pair

Compute $r(\text{Zone_Temp_F, Cooling_kWh})$ for all 8,760 hours.

The weak pair. Zone temperature barely moves ($\sigma \approx 2.9^\circ\text{F}$), so it says little about how hard the plant is working; outdoor temperature ($\sigma \approx 16.9^\circ\text{F}$) is the real driver.

A4 (MC · 2 pts) — Largest Absolute Correlation

Which correlation from A1–A3 has the largest absolute value?

(A) $r(\text{Zone_Temp_F, Outdoor_Temp_F}) = 0.7529$ (B) $r(\text{Outdoor_Temp_F, Cooling_kWh}) = 0.7752$
(C) $r(\text{Zone_Temp_F, Cooling_kWh}) = 0.2764$ (D) All three are equal — CORREL always returns the same value

A5 (2 pts) — r^2 for Outdoor Temp and Cooling

Compute $r^2(\text{Outdoor_Temp_F, Cooling_kWh})$ — the coefficient of determination.

r^2 gives the percentage of variance in Cooling_kWh linearly explained by Outdoor_Temp_F across all 8,760 hours. The remaining unexplained variance is driven by other factors — occupancy, internal equipment loads, solar gain, humidity, and HVAC scheduling.

A6 (2 pts) — $r(\text{Outdoor Temp, Cooling})$: Business Hours Only

Compute $r(\text{Outdoor_Temp_F, Cooling_kWh})$ for **business hours only** (BusinessHours = “Yes”).

Note: During occupied hours, the outdoor-cooling relationship strengthens substantially compared to the all-hours result.

A7 (MC · 2 pts) — Interpreting $r \approx 0.78$

$r(\text{Outdoor_Temp_F, Cooling_kWh}) \approx 0.78$ for all hours. Which interpretation is correct?

(A) Outdoor temperature causes cooling load. (B) There is a strong positive linear relationship — hotter outdoor hours tend to have higher cooling loads. (C) 78% of cooling hours occurred above the outdoor mean temperature. (D) The standard deviation of Cooling_kWh equals 0.78 times the standard deviation of Outdoor_Temp_F.

Causation taxonomy (ungraded): The lecture identified three reasons correlation does not equal causation: confounders (a third variable drives both), reverse causation (Y causes X rather than X causing Y), and coincidence. For this pair, outdoor temperature is a plausible physical driver of cooling load — but Season is also a confounder that independently drives both high outdoor temperatures and high cooling demand. Whenever you report a correlation, state which of the three pitfalls you have ruled out and why.

A8 (MC · 2 pts) — Why the Zone–Cooling Correlation Is Weak

$r(\text{Zone_Temp_F, Cooling_kWh}) \approx 0.28$ from A3 — much lower than $r(\text{Outdoor_Temp_F, Cooling_kWh}) \approx 0.78$ from A2. The most likely engineering explanation is:

(A) Outdoor temperature has no effect on zone temperature. (B) Cooling energy is measured in kWh, which is incompatible with °F in CORREL. (C) Zone temperature is tightly controlled near the setpoint ($\sigma_{\text{zone}} \approx 2.9^\circ\text{F}$), so it contains little information about the cooling load — all the variation in cooling demand shows up in outdoor temperature, not in the nearly-constant zone temperature. (D) Zone temperature and cooling energy are negatively correlated.

Section B Core — Seasonal and Conditional Correlations (B1–B6)

Section B shows that the single number was an average of several. Split the year by season, by occupancy, and by whether cooling ran at all, and the same relationship takes different values.

All r values → 4 decimal places.

B1 ★ (5 pts) — Seasonal r: Summer

Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for **Summer** hours only.

Col D: number of Summer hours.

B2 (4 pts) — Seasonal r: Winter

Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for **Winter** hours only.

Note: Winter still shows a moderate positive correlation. Even in winter, on warmer winter days (50–65°F), some cooling is needed — primarily during business hours when internal gains from equipment and people build up.

B3 (4 pts) — Seasonal r: Spring

Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for **Spring** hours only.

Spring has the strongest outdoor-cooling correlation of the four seasons.

B4 (4 pts) — Seasonal r: Fall

Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for **Fall** hours only.

B5 (4 pts) — Range of the Four Seasonal Correlations

Compute $\text{MAX}(B1, B2, B3, B4) - \text{MIN}(B1, B2, B3, B4)$.

This small range shows that the outdoor-cooling linear relationship is remarkably consistent across all four seasons. The relationship is strong year-round — not just in summer.

B6 ★ (5 pts) — Covariance of Outdoor Temp and Cooling

Compute $\text{COVARIANCE.S}(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for all 8,760 hours.

Col D: $\sigma(\text{Cooling_kWh})$ using STDEV.

Covariance has units ($^\circ\text{F} \times \text{kWh}$); correlation is dimensionless. Verify: $\text{cov} / (\sigma_{\text{outdoor}} \times \sigma_{\text{cooling}})$ should equal your r from A2 (this verification is stated for understanding; it is not a separate graded answer).

Engineering Context: ASHRAE 90.1-2022 requires cooling equipment sized for peak load. Correlation analysis identifies which variables are dominant drivers of that peak — informing whether a simple or complex sizing model is appropriate.

Section B Extension — Business Hours, r^2 , and Model Diagnostics (B7–B14)

B7 ★ (5 pts) — $r(\text{Outdoor Temp, Cooling})$: Business Hours

Compute $r(\text{Outdoor_Temp_F, Cooling_kWh})$ for **business hours** ($\text{BusinessHours} = \text{"Yes"}$).

Col D: number of business hours.

Note: This computes the same quantity as A6. During business hours, outdoor temperature is an almost perfect linear predictor of cooling load.

B8 (4 pts) — $r(\text{Outdoor Temp, Cooling})$: Non-Business Hours

Compute $r(\text{Outdoor_Temp_F, Cooling_kWh})$ for **non-business hours** ($\text{BusinessHours} = \text{"No"}$).

During non-business hours, cooling is reduced (setback mode or off entirely) so the outdoor-cooling relationship weakens.

B9 (4 pts) — r^2 for Business Hours

Square your B7 answer.

During occupied hours, outdoor temperature explains a very high proportion of the variance in cooling energy. The remaining unexplained variance represents other factors: solar gain variability, differences in internal loads from day to day, and HVAC control dynamics.

B10 (4 pts) — $r(\text{Zone Temp, Outdoor Temp})$: Business Hours

Compute $r(\text{Zone_Temp_F, Outdoor_Temp_F})$ for **business hours** only.

Higher than A1, which surprises people: during occupied hours the HVAC tracks outdoor conditions dynamically, so the zone moves with them in a structured way.

B11 (4 pts) — $r(\text{Month, Outdoor Temp})$

Compute $r(\text{Month, Outdoor_Temp_F})$ for all 8,760 hours.

Month is numeric 1–12, and Tucson's temperature curve is parabolic — up to July, then down. A linear correlation cannot see a rise-then-fall.

B12 (4 pts) — $r(\text{Month, Cooling})$

Compute $r(\text{Month, Cooling_kWh})$ for all 8,760 hours.

Lower still. Month is a calendar label; outdoor temperature is the mechanism. Regress on the mechanism.

B13 ★ (5 pts) — r(Outdoor Temp, Cooling): Non-Zero Cooling Hours

Compute $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ for **non-zero cooling hours** only ($\text{Cooling_kWh} > 0$).

Col D: hours where $\text{Cooling_kWh} > 0$ (COUNTIF).

Slightly higher than the all-hours r: the zero-cooling hours are cold nights and winter, anchoring both variables at their low ends.

B14 (4 pts) — Covariance of Zone Temp and Outdoor Temp

Compute $\text{COVARIANCE.S}(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$ for all 8,760 hours.

Much smaller than B6 — not a weaker relationship (the r values are close) but a smaller scale: $\sigma_{\text{zone}} \approx 2.9^\circ\text{F}$ against $\sigma_{\text{cooling}} \approx 25.3 \text{ kWh}$. Covariance carries units; correlation does not.

Required Charts (15 pts)

Both charts belong on the **Charts sheet** of your Answer Sheet, in the outlined boxes. That sheet lists what each chart is graded on.

Build the chart wherever your data is, then right-click it -> **Move Chart** -> **Object in** -> Charts, and drag it into the matching box. Leave it as a live chart — do not replace it with a picture of itself. Your preceptor clicks the chart to see which cells it plots, which is how a correct chart built from the wrong column gets caught.

Chart 1 (7 pts) — Scatter: Outdoor Temperature vs. Cooling Energy

Plot Outdoor_Temp_F (x) vs. Cooling_kWh (y) for all 8,760 rows as dots only (Insert → Scatter → dots only). Add a linear trendline with “Display R-squared value on chart” checked.

- Title: Outdoor Temperature vs. Cooling Energy – Tucson TMY3 (8,760 hours)
- Axes: Outdoor Temperature ($^\circ\text{F}$) on x, Cooling Energy (kWh) on y
- The R^2 on the chart should match your A5 answer (± 0.01)

Chart 2 (8 pts) — Seasonal Scatter: Outdoor Temperature vs. Cooling Energy

Four series on one chart, one per season, in different colours.

Build the four series on your Data sheet, in the first four empty columns after the data. Head them Summer_kWh, Winter_kWh, Spring_kWh and Fall_kWh, and in each put a formula that keeps that season’s cooling value and blanks the rest — for the Summer column, $=\text{IF}(\$E2=\text{"Summer"}, \$I2, \text{NA}())$ filled down. Plotting $\text{NA}()$ rather than 0 leaves the other seasons out of that series instead of pinning them to the axis.

Then make one scatter with Outdoor_Temp_F on the x-axis and those four columns as four y-series. Label each series in the legend with the season name and its r value from B1–B4.

- Title: Seasonal Correlation: Outdoor Temperature vs. Cooling Energy
 - Axes: Outdoor Temperature ($^\circ\text{F}$) on x, Cooling Energy (kWh) on y
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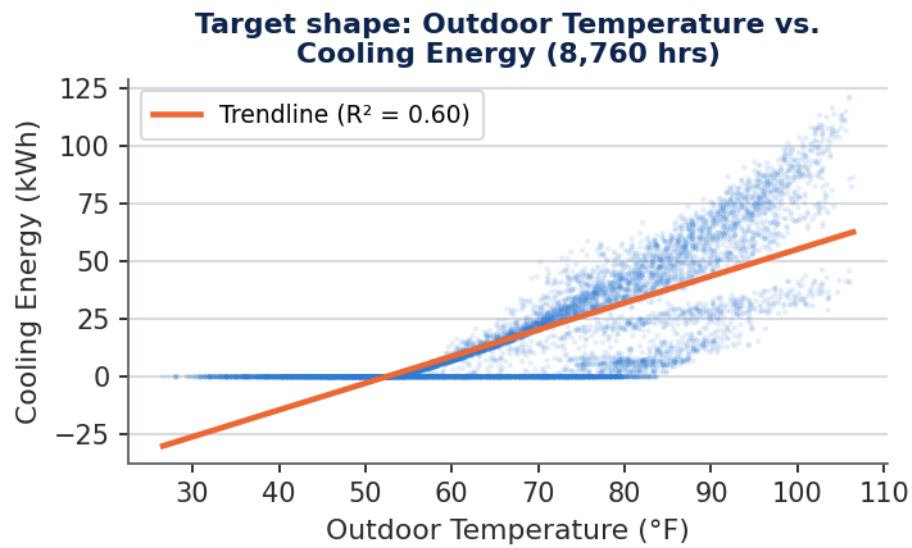


Figure 1: Target shape for Chart 1

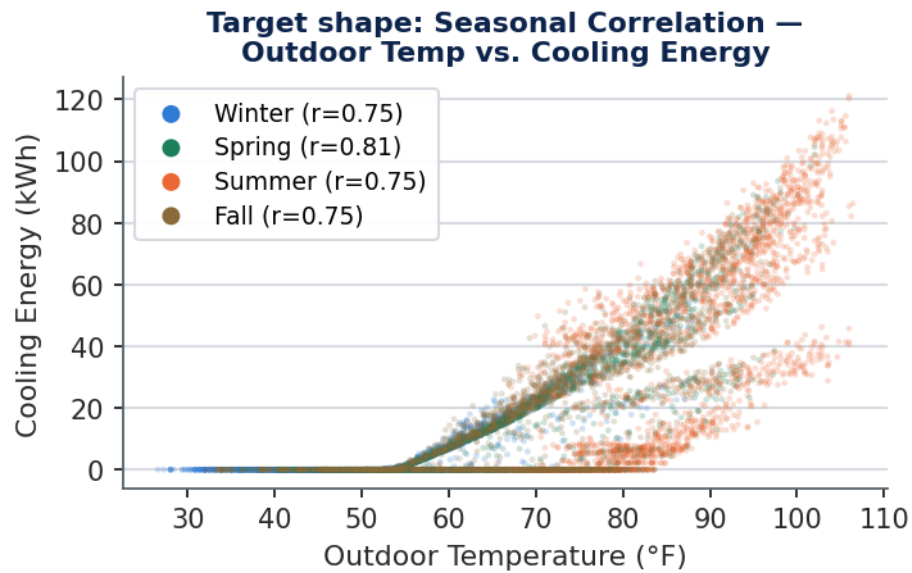


Figure 2: Target shape for Chart 2

Memo Deliverable (9 pts)

Type your memo directly in the **Written sheet** of your Answer Sheet Excel file. The sheet has labeled yellow cells for each paragraph (P1, P2, P3). Press Alt+Enter within a cell to add new lines. No separate PDF file is needed. Address the memo to: “Building Operations Team, DOE Reference Medium Office Building, Tucson AZ.”

Paragraph 1 (3 pts): Report your computed values for A1, A2, and A3. Which pair has the strongest correlation? Then explain why $r(\text{Zone_Temp_F}, \text{Cooling_kWh})$ is the weakest of the three. Compute σ for both Zone_Temp_F and Outdoor_Temp_F and cite the ratio; your explanation must say what that ratio implies about how much a controlled variable can tell you about the system controlling it.

Paragraph 2 (3 pts): Report the four seasonal r values from B1–B4 and your B5 range. Which season is strongest, and how much do the four actually differ? Referencing the ASHRAE 90.1-2022 peak-load sizing requirement in the Engineering Context note above, say what your B5 range means for equipment sizing — and be specific about whether a range that size argues for one model or four.

Paragraph 3 (3 pts) — ArcE focus: Report your computed values for B7, B8, and B9. How does the outdoor-cooling correlation differ between business hours and non-business hours? What does your $r^2(\text{business hours})$ result tell an architect about occupied-mode cooling energy? What does the remaining $(1 - r^2)$ represent, and what design factors might explain it?

Submission: Save as CE310_W7_<NetID>.xlsx and upload to the Week 7 D2L dropbox. Enter your NetID in the **NetID** cell at the top of the Answers sheet — that cell is what matches your work to your student record. A 5-point deduction applies if it is blank.

Before You Submit

- ☐ Full name and NetID entered on the Answer Sheet — a blank NetID cell takes a 5-point deduction
 - ☐ File saved as .xlsx and named CE310_W7_<NetID>.xlsx
 - ☐ Sheet tabs not renamed, reordered, or deleted
 - ☐ Every answer typed into its designated cell in **Col E** — answers placed anywhere else receive no credit
 - ☐ Your formula typed as text in **Col C**, with an apostrophe before the = — screenshots are not accepted
 - ☐ ★ questions also have their intermediate value in **Col D**
 - ☐ Both charts sitting in their boxes on the Charts sheet, as live charts and not pictures, with titles and axis labels
 - ☐ Memo written in the Written sheet
 - ☐ Uploaded to D2L by **9:00 AM next Wednesday**
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Grading Summary

Component	Points
Section A (A1–A8)	16

Component	Points
Section B (B1–B14)	60
Chart 1 (scatter + trendline)	7
Chart 2 (seasonal scatter)	8
Memo (3 paragraphs)	9
Total	100

Point structure: Section A = 8×2.0 pts (0.5 approach + 1.5 answer each). Section B = 10 standard $\times 4.0$ pts + 4 intermediate $\star \times 5.0$ pts = $40 + 20 = 60$ pts. $\star$ questions: B1, B6, B7, B13.